

CSA15 Cloud Computing and Big Data Analytics

CO4 AT1 - Big Data Analytics Class Test Student Template

Course Code	CSA15	Slot	B
Course Title	Cloud Computing and Big Data Analytics	Assessment	CO4 AT1
Assessment Title	Big Data Analytics Class Test	Bloom Level	BL4 - Analyze
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Submission Mode	Individual digital answer template through GitHub and Google Classroom	Total Marks	30

Student Details

Student Name	Konatham Mahesh	Register Number	192311014
Class / Section		Slot	B
Capstone Title	SafeWalk – Cloud Route Intelligence Using Crowd Reports and Risk Heatmaps	Submission Date	19/08/26
GitHub Repository Link	https://github.com/konatham2005/CSA1512	Google Classroom Status	https://classroom.google.com/u/1/c/ODcwMzQ3MDExMjg1/a/ODU1ODg3NjM4Njg1/details

Assessment Overview

CO4 Focus

Analyze big data characteristics, processing needs, security challenges, business drivers and domain applications. This template includes explanation notes to guide the thinking process. The final answer must be written in the student's own words.

How to use this template:

- Read each question and the explanation notes before answering.
- Use your capstone product as an example wherever relevant.
- Write concise technical answers; avoid copying the explanation notes directly.
- Use diagrams or small tables wherever they improve clarity.
- Upload the completed DOCX/PDF and supporting evidence to GitHub; share the repository link through Google Classroom.

Assessment Rubric

Criteria	Marks	Evaluation Focus	Student Check
Concept Accuracy	9	Correct technical meaning, terminology and examples.	Have I used correct big data terms?

Coverage	6	All required parts of the question are answered.	Have I answered every part?
Examples	4.5	Relevant cloud/capstone/domain examples are included.	Have I used meaningful examples?
Analysis	6	Implications, security, scalability and decision value are explained.	Have I explained why it matters?
Clarity	4.5	Answer is organized, concise and readable.	Is my answer easy to evaluate?

Code of Conduct and Student Assurance

I Konatham mahesh / 192311014 (Name / Register Number) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: konatham mahesh

Question and Answer Sections

Answer all questions. Each question carries 3 marks. Use the explanation notes for direction, but write the final answer independently.

Question 1 (3 Marks)

Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

Explanation Notes

1	Start with a precise definition: Big Data refers to datasets whose size, speed, variety or complexity exceeds the capability of conventional tools.
2	Mention why spreadsheets or single-server databases struggle: storage limits, slow queries, concurrency issues, schema rigidity and limited scalability.
3	Use one cloud/capstone-style example such as app logs, IoT feeds, transactions, images or clickstream events.
4	End with why distributed storage and parallel processing become necessary.

Student Response

- Big Data refers to datasets that are very large, fast-changing, diverse, or complex and cannot be efficiently handled by traditional data-processing methods. In SafeWalk, data such as crowd safety reports, route information, location data, and risk records can grow rapidly.
- Traditional spreadsheets or single-server databases may have storage, processing, concurrency, and scalability limitations. Therefore, SafeWalk can use cloud-based distributed storage and parallel processing to manage and analyze large amounts of data efficiently.

Question 2 (3 Marks)

Explain the 5Vs of Big Data with one practical example for any two Vs.

Explanation Notes

1	List the 5Vs clearly: Volume, Velocity, Variety, Veracity and Value.
2	For at least two Vs, connect the definition to a concrete scenario instead of writing only textbook meaning.
3	Example: In a smart traffic system, velocity refers to continuous vehicle movement updates; value refers to congestion prediction.
4	Show that each V affects system design, storage, processing or analytics decisions.

Student Response

- The 5Vs of Big Data are **Volume, Velocity, Variety, Veracity, and Value**.
- In SafeWalk, **Velocity** represents the continuous arrival of crowd reports and route updates. **Variety** represents different types of information such as route records, location data, text reports, and map-related data. These characteristics influence how the system stores and processes data in the cloud.

Question 3 (3 Marks)

Differentiate structured, semi-structured, and unstructured data with examples.

Explanation Notes

1	Structured data has a fixed schema, such as customer table, transaction table or attendance record.
2	Semi-structured data has tags/keys but flexible structure, such as JSON, XML, API responses or log files.
3	Unstructured data has no fixed tabular form, such as images, videos, audio, PDFs, emails or social media text.
4	Use examples from one cloud-backed application to make the comparison coherent.

Student Response

- Structured data is organized in a fixed format, such as user ID, route ID, timestamp, and risk score in SafeWalk. Semi-structured data has a flexible structure with keys or tags, such as JSON-based route and crowd-report data. Unstructured data does not follow a fixed tabular format, such as user comments, incident descriptions, photographs, or other report content. SafeWalk can combine these different data types to analyze routes and identify potentially risky areas.

Question 4 (3 Marks)

Explain the role of distributed file systems in big data processing.

Explanation Notes

1	State the main need: very large datasets are split and stored across multiple machines.
2	Mention core benefits: scalability, fault tolerance, parallel access and support for batch processing.
3	Use HDFS or cloud object storage as an example, but do not confuse file storage with relational database storage.
4	Connect storage to processing: data blocks can be processed in parallel by frameworks such as MapReduce or Spark.

Student Response

- A distributed file system divides large datasets into smaller parts and stores them across multiple machines. In SafeWalk, large amounts of crowd reports, route histories, and location-related data can be stored using distributed cloud storage. The data can then be processed in parallel using frameworks such as MapReduce or Spark. This provides scalability, fault tolerance, and efficient processing of large datasets.

Question 5 (3 Marks)

List any three major security concerns in big data platforms and briefly explain each.

Explanation Notes

1	Choose three concerns such as unauthorized access, data leakage, privacy violation, weak encryption, poor audit logs or insecure APIs.
2	For each concern, explain the risk and not only the keyword.
3	Add one control where possible: role-based access, encryption, anonymization, masking, logging or monitoring.
4	Show awareness that big data combines many data sources, so security and governance become more complex.

Student Response

- Three major security concerns in SafeWalk are unauthorized access, data leakage, and privacy violation. Unauthorized access may allow unwanted users to view route or crowd-report information, which can be controlled using role-based access. Data leakage may expose sensitive location information, so encryption should be used for stored and transmitted data. Privacy violation may occur if user movement information reveals personal details, so anonymization and data masking can be applied to protect user privacy.

Question 6 (3 Marks)

What are data appliances and integration tools? Explain their role in big data workflows.

Explanation Notes

1	A data appliance is a specialized hardware/software platform optimized for storage, processing or analytics workloads.
2	Integration tools move, transform or synchronize data between sources, storage systems and analytics layers.
3	Mention ETL/ELT, API connectors, message queues, ingestion tools or workflow schedulers as examples.
4	Explain the workflow: source data -> ingestion -> cleaning/transformation -> storage -> processing -> dashboard/decision.

Student Response

- A data appliance is a specialized hardware or software platform designed for storage, processing, or analytics workloads. Integration tools help move, transform, and synchronize data between different systems. In SafeWalk, data from users and other sources can be collected, cleaned, transformed, and stored in the cloud before analytics is performed. The workflow can be represented as data sources → ingestion → cleaning and transformation → cloud storage → processing and analytics → risk heatmap and route recommendation.

Question 7 (3 Marks)

Discuss two major challenges organizations face while implementing big data analytics.

Explanation Notes

1	Pick two strong challenges such as data quality, skill gap, infrastructure cost, privacy compliance, integration difficulty or real-time scalability.
2	Explain the cause and impact of each challenge.
3	Suggest a practical mitigation for each challenge, such as governance, cloud scaling, training, access control or phased implementation.
4	Try to connect the challenge to business impact, not only technical difficulty.

Student Response

- Two major challenges in implementing Big Data Analytics for SafeWalk are data quality and real-time scalability. Crowd reports may contain incorrect, duplicate, incomplete, or unreliable information, which can affect the accuracy of risk analysis. Data validation and filtering can help improve data quality. The system may also receive a large number of reports and route updates, requiring real-time scalability. Cloud scaling and distributed processing can help SafeWalk handle increasing workloads efficiently.

Question 8 (3 Marks)

Write short notes on any one domain application of big data analytics.

Explanation Notes

1	Select one domain: healthcare, retail, agriculture, finance, education, smart city, transport or cybersecurity.
2	Identify data sources in that domain.
3	Explain how analytics converts raw data into decisions: prediction, recommendation, alert, optimization or risk detection.
4	Mention one expected outcome such as reduced waiting time, fraud detection, crop advisory or improved user experience.

Student Response

- SafeWalk can be considered a smart city application of Big Data Analytics. It can collect crowd reports, route information, location data, and historical risk information from different sources. Analytics can identify patterns in safety reports and generate risk heatmaps for different areas. These results can be used to recommend safer routes instead of relying only on the shortest route. This can improve user safety and support better movement decisions in urban areas.

Question 9 (3 Marks)

How do business drivers influence the adoption of big data solutions?

Explanation Notes

1	Business drivers are reasons that push an organization to adopt big data, such as cost reduction, personalization, speed, risk control or better decisions.
2	Explain how each selected driver creates a technical requirement.
3	Example: personalization requires behavior tracking and recommendation analytics; risk control requires fraud detection and monitoring.
4	Conclude that big data adoption must be justified by measurable business value.

Student Response

Business drivers are the reasons that encourage an organization to adopt Big Data solutions. For SafeWalk, important business drivers include risk control, better decision making, faster services, and improved user experience. Risk control requires analyzing crowd reports and historical data to identify potentially unsafe areas. The analysis can then be converted into risk scores and safer route recommendations. Therefore, Big Data adoption should provide measurable value such as improved safety and user trust.

Question 10 (3 Marks)

Explain the importance of big data analytics in decision making.

Explanation Notes

1	Explain that analytics helps convert raw, high-volume data into useful patterns, predictions, alerts or recommendations.
2	Differentiate descriptive, diagnostic, predictive and prescriptive decision support if space permits.
3	Use one example: dashboard for hospital beds, forecast for demand, fraud alert for bank, traffic prediction for smart city.
4	End with the idea that decisions become faster, evidence-based and scalable.

Student Response

➤ Big Data Analytics is important for decision making because it converts large amounts of raw data into useful patterns, predictions, alerts, and recommendations. In SafeWalk, analytics can examine crowd reports, historical route information, and risk-related data to identify potentially unsafe areas. The results can be displayed through risk heatmaps and used to recommend safer routes. Therefore, Big Data Analytics helps SafeWalk provide faster, evidence-based, and scalable route safety decisions.

Submission Checklist

Checklist Item	Status	Remarks
Student details completed.	Done / Pending	NA
All 10 questions answered.	Done / Pending	NA
Examples are connected to cloud/capstone/domain context.	Done / Pending	NA
Security/privacy points are included wherever relevant.	Done / Pending	NA
Document exported as PDF if requested.	Done / Pending	NA
GitHub link shared through Google Classroom.	Done / Pending	NA